MIL-C-43247B

16 September 1970

SUPERSEDING
MIL-C-43247A
17 March 1967

MILITARY SPECIFICATION

CLOTH, KNITTED, NYLON, TUBULAR, STRETCH TYPE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for three classes of knitted nylon stretch type tubular cloth.

1.2 Classification.- The cloth shall be of the following classes, as specified (see 6.2):

Class 1 - 7.0 ± 0.5 ounces per square yard (interlock)
Class 2 - 10.5 ± 0.5 ounces per square yard (interlock)
Class 3 - 10.0-11.5 ounces per square yard (1x1 rib knit)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

FSC 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample.- The cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.-

* 3.2.1 Yarn.- The circular knitted nylon cloth shall be made from a stretch type nylon yarn (see 6.5).

3.3 Color.-

3.3.1 Classes 1 and 2.- The color for classes 1 and 2 finished cloth shall be Olive Green 106 (see 6.4).

3.3.2 Class 3.- The color for class 3 finished cloth shall be Blue 3328.

3.3.3 Matching.- The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.4 Colorfastness.- The finished cloth shall show fastness to laundering, crocking and perspiration equal to or better than the standard sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

When no standard sample is established, the cloth shall show "good" colorfastness to laundering and perspiration and a Munsell Value for crocking not lower than 8.5 when tested as specified in 4.4.

3.4 Physical requirements.- The finished cloth shall conform to the physical requirements specified in table I when tested as specified in 4.4.

TABLE I.- Physical requirements

Class	Knit type	Weight oz/sq.yd.	Wales per inch (min)	Courses per inch (min)	Bursting strength pounds (min)	Elongation - percentage length (wales) (min)	(max)	width (courses) (min)	(max)
1	Interlock	7.0 ± 0.5	30	35	165	70	125	120	175
2	Interlock	10.5 ± 0.5	28	40	200	90	115	135	160
3	1x1 rib	10.0 min. 11.5 max.	15	28	-	60	90	200	300

3.4.1 Width.- The minimum width of the slit and finished cloth shall be as specified (see 6.2).

3.5 Finish.- The tubular cloth shall be scoured, dyed, slit, and heat treated.

3.5.1 pH.- The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.

3.6 Dimensional stability (classes 1 and 2).- The finished cloth shall not shrink or elongate more than the percentages shown in table II when tested as specified in 4.4.

TABLE II.- Dimensional stability

	Length (wales) (maximum)	Width (courses) (maximum)
Shrinkage (percent)	2.0	3.0
Elongation (percent)	2.0	3.0

3.7 Length and put-up.- Unless otherwise specified (see 6.2), the finished cloth shall be in continuous lengths of not less than 20 yards and shall be put up on rolls as specified in PPP-P-1133.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

3.8 Fiber identification.- Each roll of cloth shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.9 Workmanship.- The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents.

4.2.2 Examination of the end item.- Defects found during this examination shall be classified in accordance with 4.2.2.1, 4.2.2.2 and 4.2.2.3.

4.2.2.1 Yard-by-yard examination.- The required yardage shall be inspected on one side and visual defects classified as listed. The side examined shall be alternated on every other roll. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each wale-wise (lengthwise) yard or fraction thereof in which it occurs. The sample unit for this examination shall be one linear yard. The acceptable quality levels for this examination shall be 1.5 major defects and 6.5 total defects (major and minor combined) per 100 units (yards). The sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

size shall be based upon inspection level II of MIL-STD-105. The lot size shall be expressed in units of one yard. The number of rolls from which the sample is to be selected shall be in accordance with table III. An approximately equal number of yards shall be examined from each roll in the sample.

Defects	Classification Major	Minor
Any hole, cut, tear	X	
Run or dropped stitch, end out	X	
Tuck stitch, pulled stitch or needle streak		X
Mend, any	X	
Abrasion mark, resulting in a weak area	X	
Thin or uneven cloth	X	
Coarse or fine yarns 1/	X	
Slubs or slugs or gouts, more than twice the size of normal yarn		X
Foreign matter 1/		X
Knots oversize, extending above the surface of cloth		X
Kinks or untrimmed ends		X
Crease or wrinkle, embedded		X
Stretch (stitch tension) knitting tension too tight or too loose	X	
Shade bar 1/		X
Spot or stain	X	

1/ Clearly visible at normal inspection distance (approximately 3 feet).

4.2.2.2 Overall examination.- Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the number of defects acceptable shall be as shown in table III.

Defects

Width less than specified.
Overall uncleanness.
Uneven knitting throughout the piece.
Off shade, shaded end to end, side to side, side to center, or throughout the roll.
Mottled, streaky or cloudy appearance.
Edges cut, frayed, curled, rolled, folded, scalloped or uneven continuously or intermittently throughout the roll.
Not labeled in accordance with the Textile Fiber Products Identification Act.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

4.2.2.3 Length examination.-

4.2.2.3.1 Individual rolls.- The rolls shall be examined for gross length. Any gross length found to be less than the minimum specified or more than 2 yards less than the gross length marked on the ticket shall be considered a defect with respect to length. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in table III.

4.2.2.3.2 Total yardage in sample.- The lot shall be unacceptable if the total of the actual gross length of rolls in the sample is less than the total of the gross lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

TABLE III.- Sample sizes

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.3 Examination of preparation for delivery requirements.- An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with the section 5 requirements of this specification.

4.4 Testing of the end item.- The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table V shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 2-1/2 yards full width of the finished cloth. The sample size shall be as listed in table IV. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

TABLE IV.- Sample size

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE V.- Test methods

Characteristic	Requirement paragraph	Test method
Nylon stretch yarn:		
Material identification	3.2.1	1/
Colorfastness to:		
Laundering	3.3.4	5610
Crocking	3.3.4	5651
Perspiration	3.3.4	5680
Knit	3.4	Visual 2/
Weight	3.4	5041
Texture:		
Wales per inch	3.4	5070
Courses per inch	3.4	5070
Bursting strength (classes 1 and 2)	3.4	5120
Elongation (percent):		
Wales	3.4	4.4.1
Courses	3.4	4.4.1
pH	3.5.1	2811
Dimensional stability (classes 1 and 2)	3.6	5556 3/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

2/ One determination shall be made on each sample unit and the result reported as "pass" or "fail".

3/ The cotton laundering procedure shall be followed.

4.4.1 Elongation.- Elongation of the cloth shall be measured on an inclined plane type of machine (IP4 or similar type) with a suitable autographic recording device being used. Specimens of the cloth shall be cut 4 by 6 inches with the long direction parallel to the direction to be tested. One-inch clamps, placed two inches apart shall be used with the specimen placed in the clamp jaws without tension or slack. The machine shall run through ten complete cycles, proceeding to its maximum depression so as to exert a five-pound tension. Five specimens each containing different wales and courses shall be tested, using individual recorded stretch graphs. The average stretch of five specimens shall be measured at the peak of last recorded cycle, in a line perpendicular to the horizontal zero axis. The percent elongation shall be calculated as follows:

$$\text{Percent elongation} = \frac{\text{Elongated dimension} - \text{original dimension}}{\text{Original dimension}} \times 100$$

4.4.1.1 Report.- Five specimens from each of the length (wale) and width (course) directions shall be tested from each sample unit. The results shall be reported as the average of the five specimens tested in each direction and reported separately to the nearest 1.0 percent.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging.- Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C.- The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

6. NOTES

6.1 Intended use.- The cloth covered by this specification is intended for use as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

(a) Class 1 cloth is intended for use in the manufacturing of man's field coats.
(b) Class 2 cloth is intended for use in the manufacturing of insulating caps for helmet liners.
(c) Class 3 cloth is intended for use in the manufacturing of utility coveralls for Navy personnel aboard nuclear type submarines.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class of cloth required (see 1.2).
(c) Width of cloth required (see 3.4.1).
(d) Length, if other than specified (see 3.7).
(e) Selection of applicable levels of put-up packaging and packing (see 5.1 and 5.2).

6.3 Standard sample.- For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Dyestuff formulation.- A suggested but not mandatory dye formulation for Olive Green 106 is as follows:

Acid Yellow 114
Neutral Premetallized Olive BGL
Neutral Premetallized Green BLS

* 6.5 Class 1 cloth was knitted from stretch type nylon yarn processed from two ends of 70 denier nylon (one end per feed). Classes 2 and 3 cloth were knitted from stretch type nylon yarn processed from two ends of 100 denier nylon (one end per feed for class 2 and two ends per feed for class 3).

6.6 The margins of this specification are marked with an asterisk to indicate where changes, (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Air Force - 82

User activity:

Navy - SH

Preparing activity:

Army - GL

Project No. 8305-0528

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-ROS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: Cloth, Knitted, Nylon, Tubular, Stretch Type MIL-C-43247B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:59Z
Check the source to verify that this is the current version before use.